

Basic Principles of Medicine 1

Module : Cardiovascular, Respiratory and Renal DLA

Development of the Cardiovascular system and Heart

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Recommended Reading

- Moore, Persaud, Torchia. The developing human. Clinically oriented embryology 11th Ed.
- Printed Textbook: Chapter 13, Pgs. 263 - 291.
- Online Textbook: Chapter 3: Development of Cardiovascular system
- Pay special attention to the **Green/ blue** boxes for the clinical correlations for this topic.

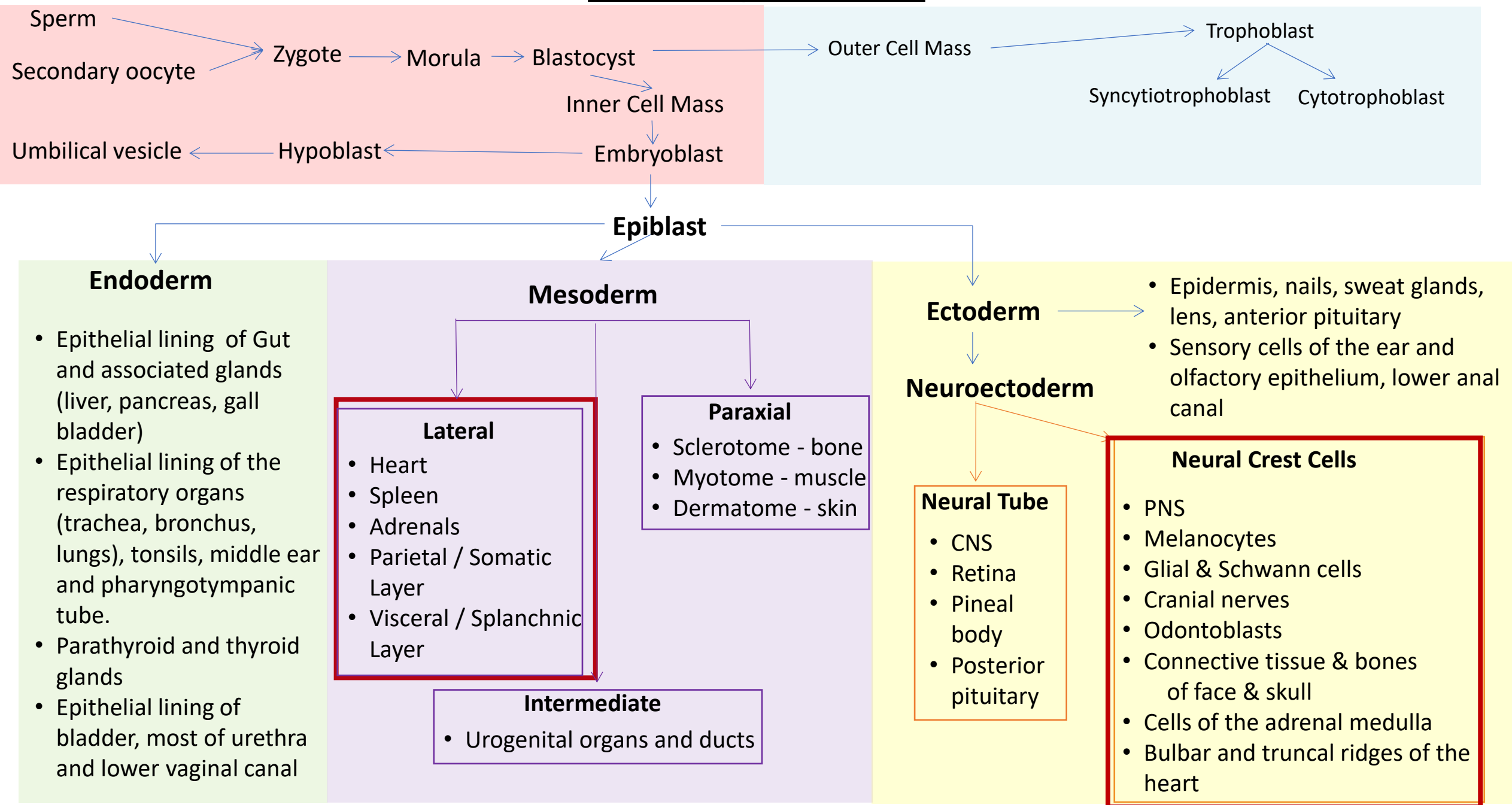
Images taken from:

- The Developing Human Clinically Oriented Embryology, 11th Edition. Keith L. Moore, T.V.N. Persaud, Mark G. Torchia. *Elsevier*
- Langman's Medical Embryology, 14th Edition. T.W. Sadler. *Wolters Kluwer*
- Peter H. Abrahams, Jonathan D. Spratt – McMinn's & Abrahams' Clinical Atlas of Human Anatomy, 7th Edition (2013, Mosby).
- Atlas of Human Anatomy, 6th Edition – Frank H. Netter, M.D.

Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1402	Describe the formation of the oblique and the transverse pericardial sinus.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1403	List the subdivisions of the tubular heart.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1404	Describe the difference between dextrocardia and situs inversus
SOM.MK.II.BPM1.3.CPR.2.ANAT.1405	Describe the partitioning of the atrioventricular canal.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1406	Describe the formation of the right and left atria.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1407	Describe the development of the interatrial septum.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1408	Describe the fate of the sinus venosus
SOM.MK.II.BPM1.3.CPR.2.ANAT.1410	Describe the partitioning of the common ventricle.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1411	Describe the partitioning of the bulbus cordis and truncus arteriosus by the aorticopulmonary septum.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1413	Describe the development of the heart valves.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1414	List the cardiac structures derived from the neural crest cells.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1415	Explain the fate of vitelline, umbilical and cardinal vessels.

The Big picture



The journey – development of the cardiovascular system

Recall the development of the notochord in week 3

- Epiblast cells migrate through the primitive streak and into the lateral plate mesoderm – Day 16 to 18
- Primary cardiogenic/heart field (PHF) is established – Day 18
- Left and right dorsal aorta are formed – Day 18-20
- Left and right endocardial tubes are formed – Day 20
- Embryo starts to fold – Day 20
- Heartbeat is initiated – day 21
- Heart starts to pump blood – day 24/25
- Endoderm induces formation of lateral endocardial tubes within the PHF
- Embryo continues to fold craniocaudally and laterally
 - Endocardial tubes fuse in the ventral midline
 - Establishes the primary heart tube
- Primary heart tube undergoes changes
 - **Looping, remodelling, realignment and septation**

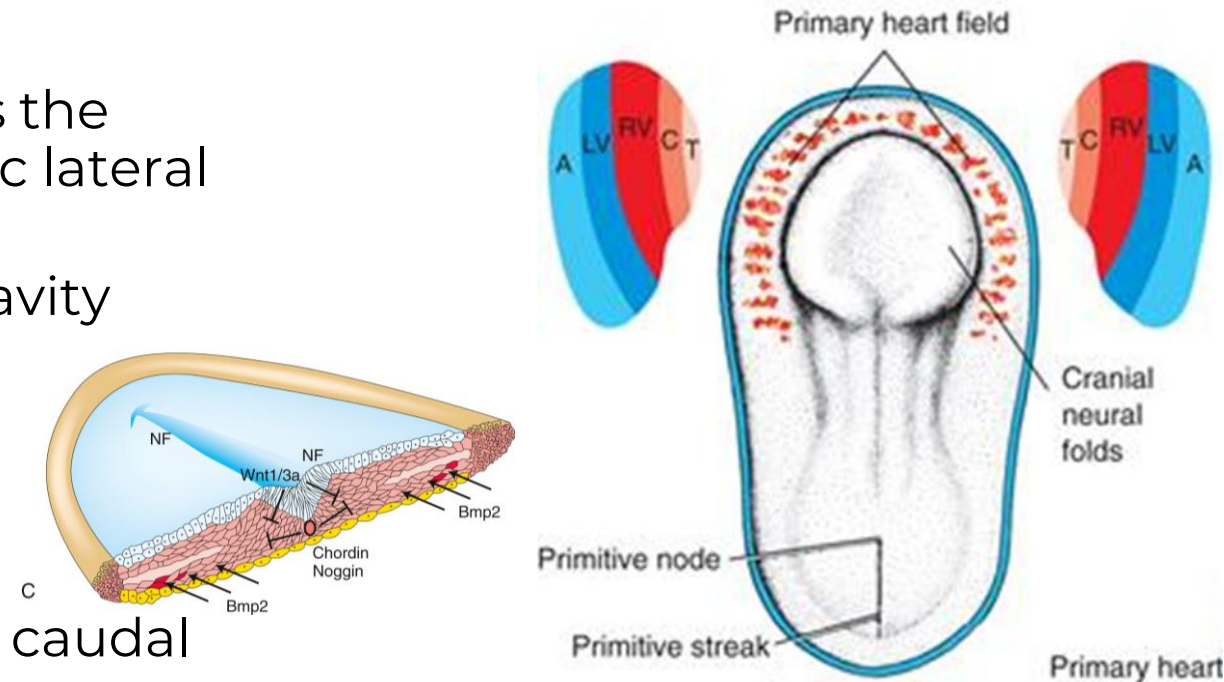
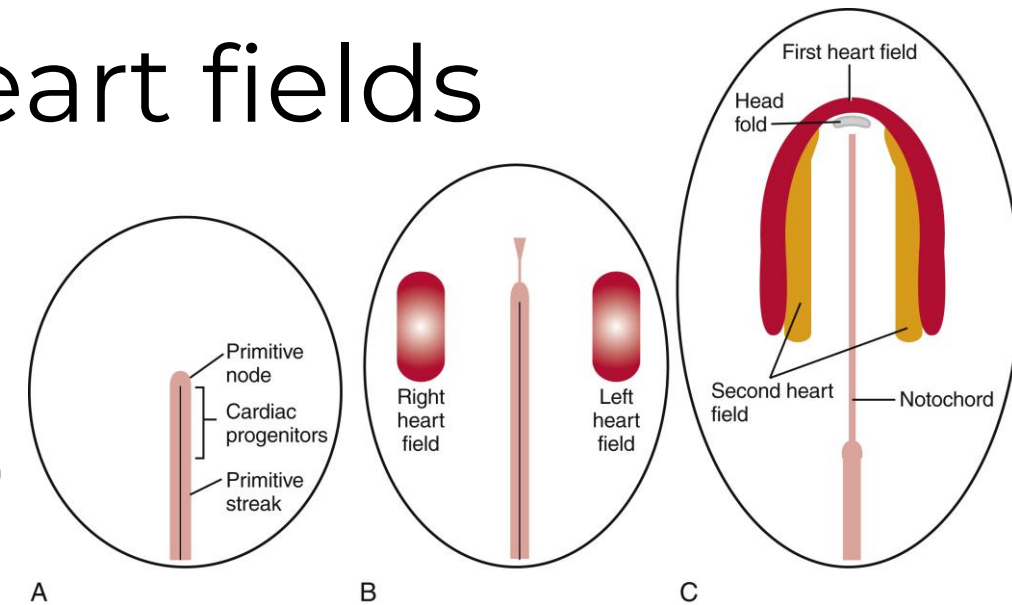
Week 3

Week 4

Weeks 4 to 8

Week 3 – Development of heart fields

- Day 16 – **Notochordal** process begins to develop
- Day 16 – Epiblast cells migrate through the **cranial third** of the primitive streak
 - Settle in the **intraembryonic mesoderm** lateral to the primitive streak (**left and right heart fields**)
 - Cell position from medial to lateral is based on the part of the heart they will contribute to
- Day 18 – **Notochord** is completely formed
- Day 18 - The intraembryonic coelom separates the splanchnic lateral mesoderm from the somatic lateral mesoderm
 - This cavity later becomes the pericardial cavity
- Day 18 - Cells continue migration as the neural folds develop forming a **Crescent-shaped Primary Heart Field** in the splanchnic lateral mesoderm
- A second heart field forms medial and slightly caudal to the PHF

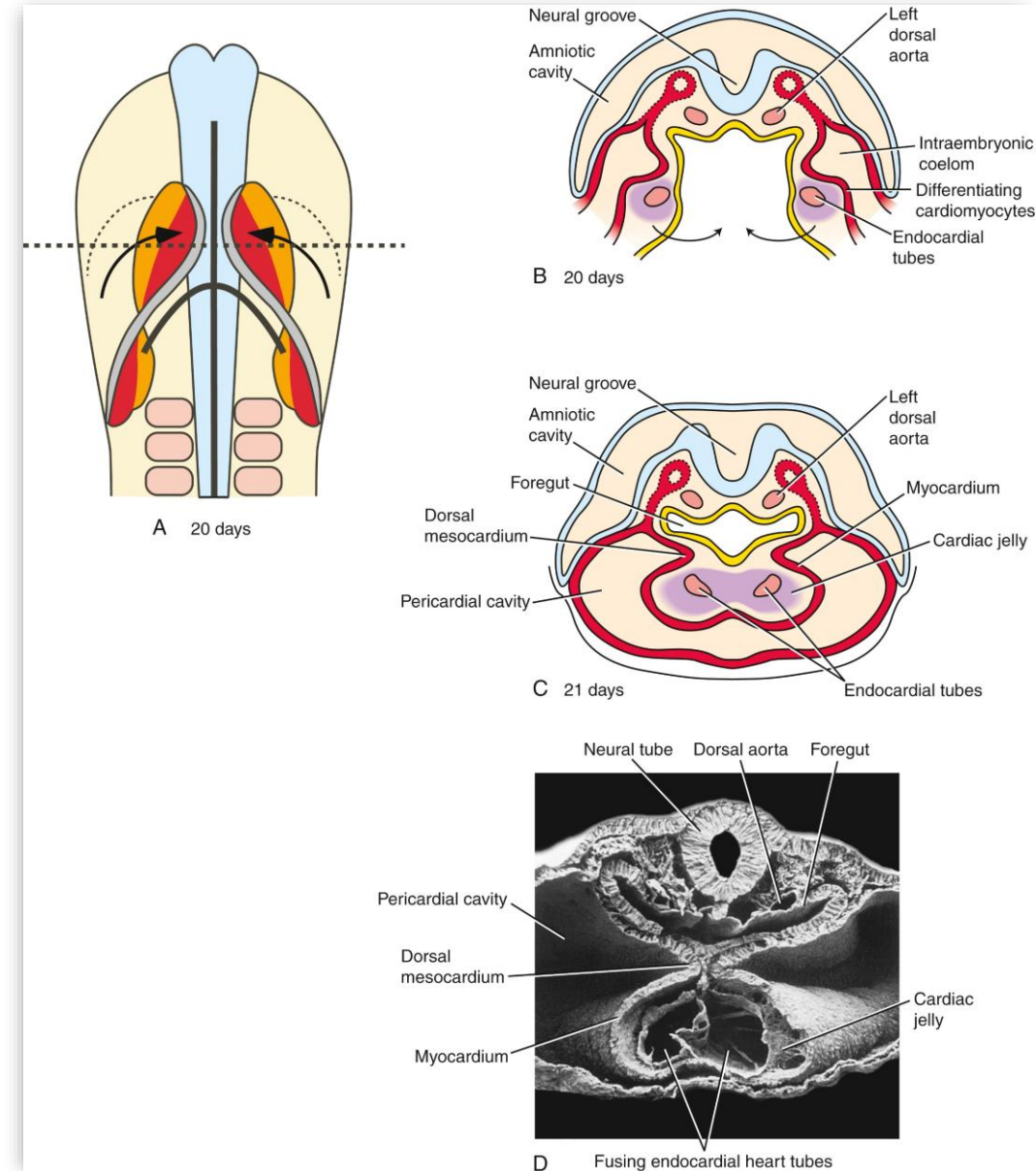


Week 3 cont.

- Day 20 - The underlying endoderm induces the cells in each of the PHF to form a tube = **endocardial tube***

the mechanism by which this happens is not understood

- Bilaterally to the notochord the **dorsal aorta** forms in the dorsal mesenchyme of the embryonic disc
- The embryo begins to fold
- The headfold
 - pulls the **cranial portion of the PHF ventrally and caudally** so it lies ventral to the developing foregut
 - the **dorsal aorta is pulled ventrally** fuses with the developing heart tube = **1st aortic arch** artery is formed
- The lateral folds
 - bring the left and right endocardial tubes **side by side**

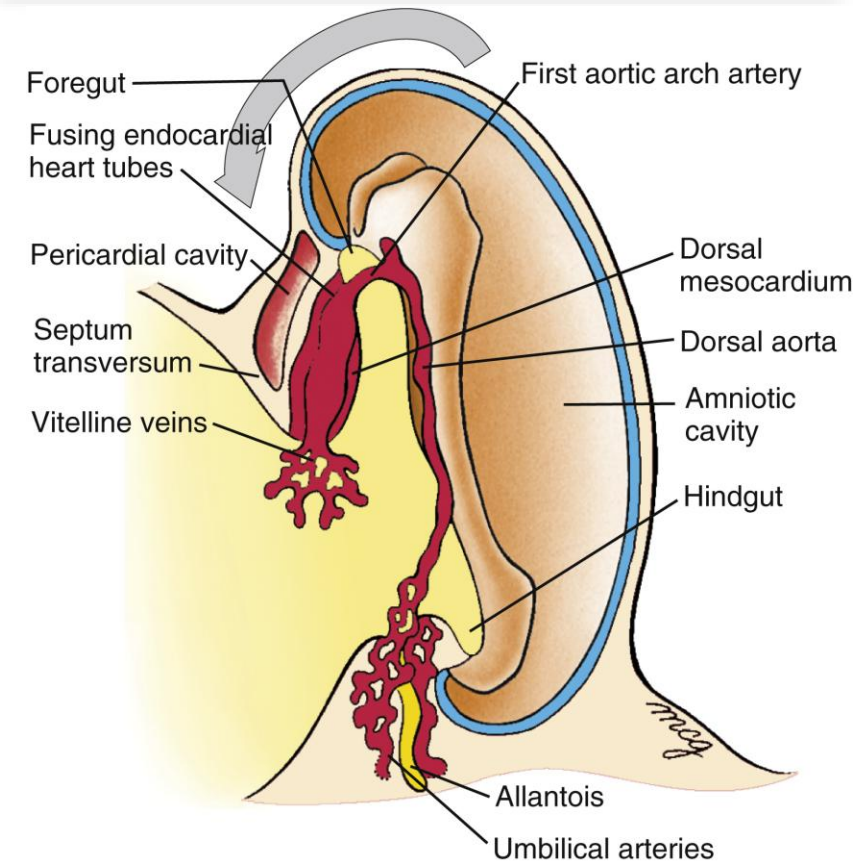
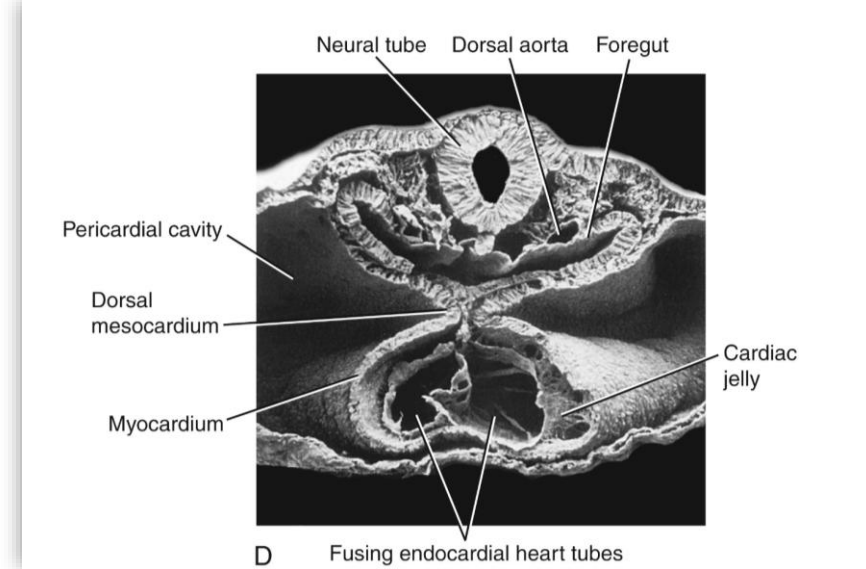


Week 3 cont. Day 21

- Cardiogenic cells start contracting*
****this contractility plays an important role in the development of the heart****
- The left and right endocardial tubes fuse to form a single heart tube
- Cell proliferation adds to the to the caudal segments of the heart
- Primary heart tube contains progenitors for the atria, left ventricle and endocardium
- Continued folding brings the developing heart tube into the thoracic region

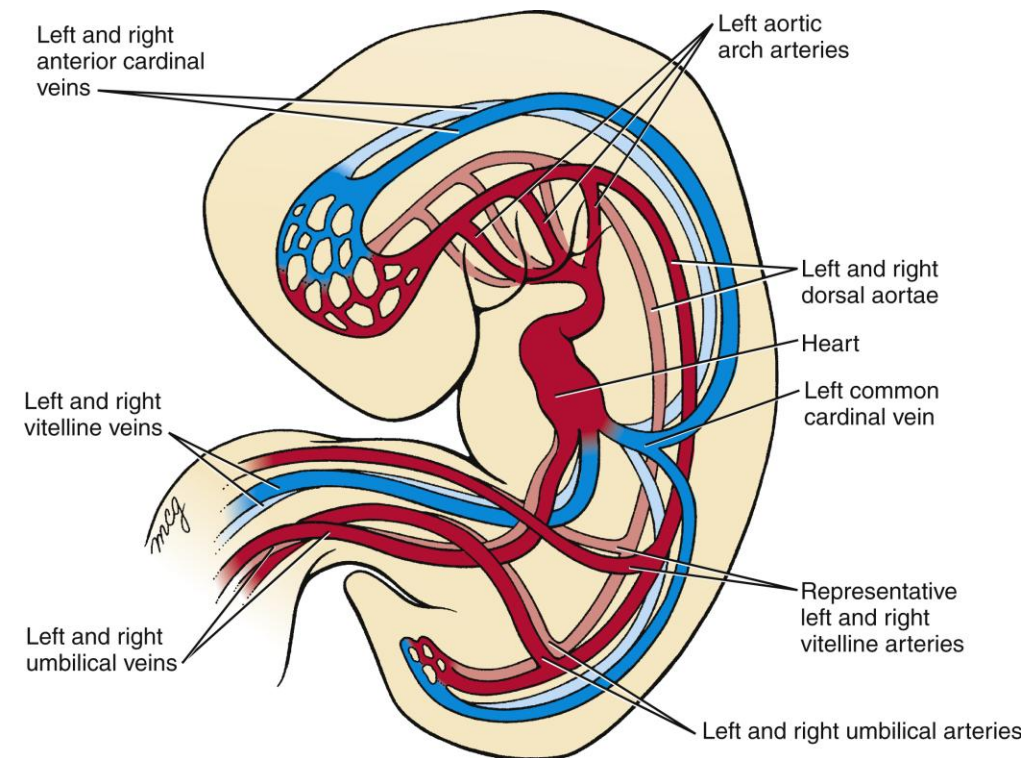


EMG showing the progression from the cardiogenic field (arrow) to formation of the primitive heart tube (B), proliferation of cardiogenic cells and repositioning of the developing heart (C)



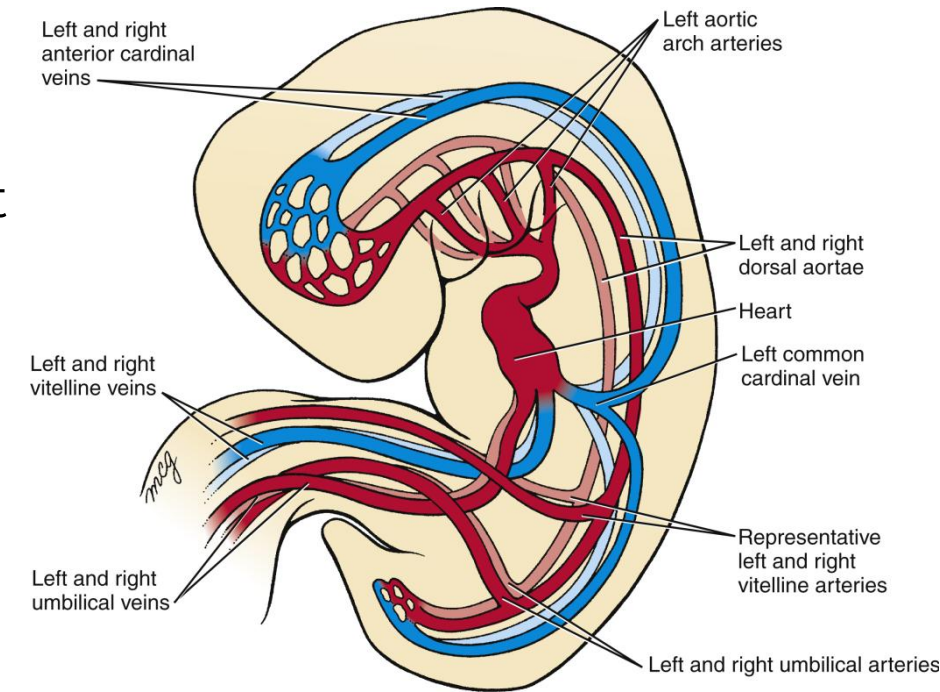
Development of the venous connections to the developing heart

- **3 venous systems** develop alongside and simultaneous to the endocardial tubes
 - Initially bilaterally symmetrical
 - Fuse with the sinus horns (the unfused caudal portion of the endocardial tubes)
- The **cardinal system** – brings blood from the body of the embryo to the developing heart
 - Anterior cardinal veins – drains cranial portion of embryo
 - Posterior cardinal veins – drains caudal portion of embryo
 - Joins to form the left and right common cardinal vein that connects with the to the sinus venosus
- **Vitelline system** – brings blood from the umbilical vesicle to the heart
 - Left vitelline vein regresses
 - Right vitelline vein forms the hepatic portal system and a portion of the inferior vena cava



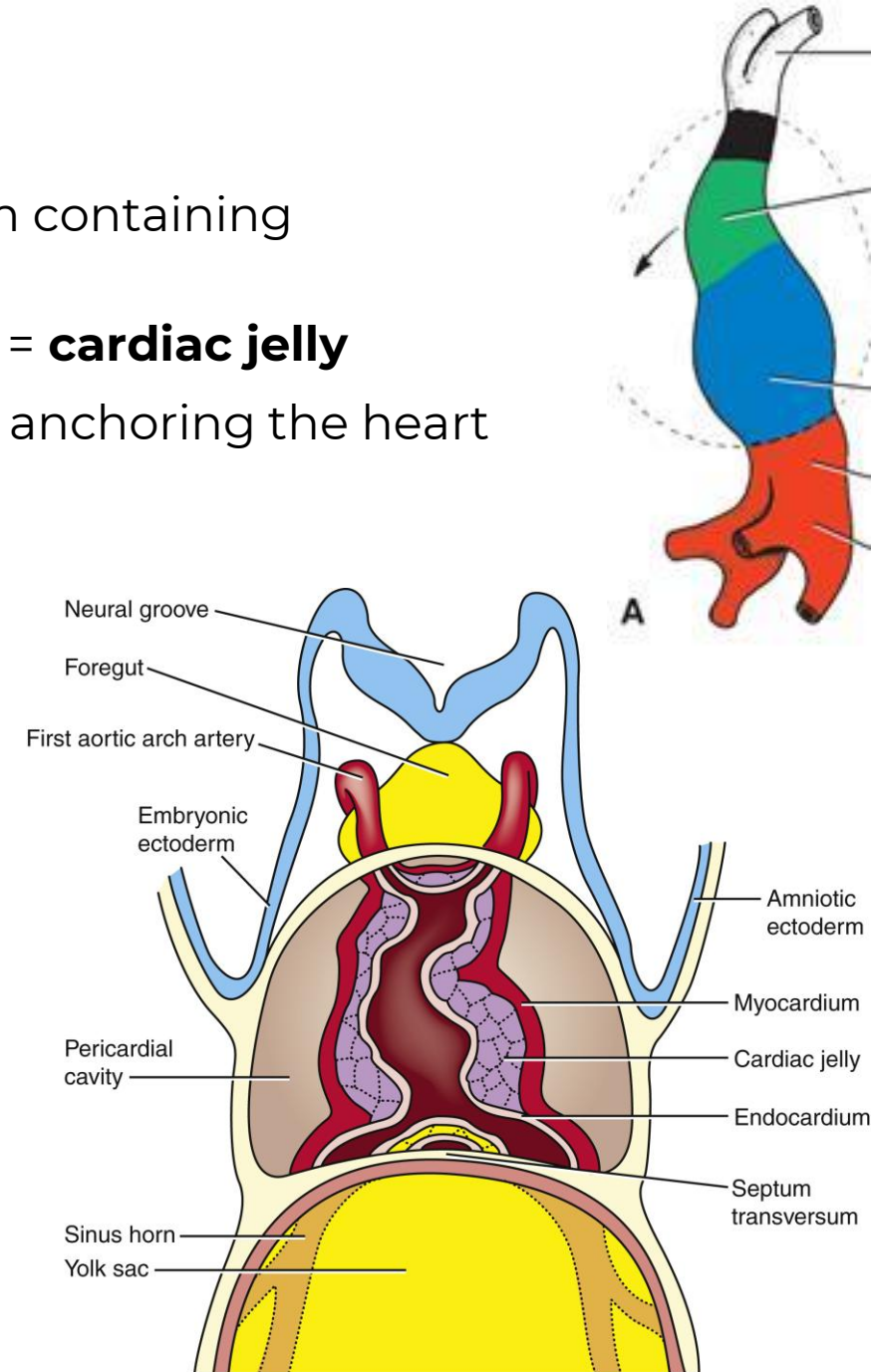
Development of the venous connections to the developing heart

- **Umbilical system** – brings oxygenated blood from the placenta
 - During development of the liver, they lose their connection with the heart and empty into the liver
 - Right umbilical vein and the cranial portion of the left umbilical vein degenerates
 - The caudal portion of the left umbilical vein becomes the persistent umbilical vein
 - Ductus venosus is a shunt that forms between the umbilical vein and the inferior vena cava allowing blood to pass directly to the heart



Week 4 – Day 22

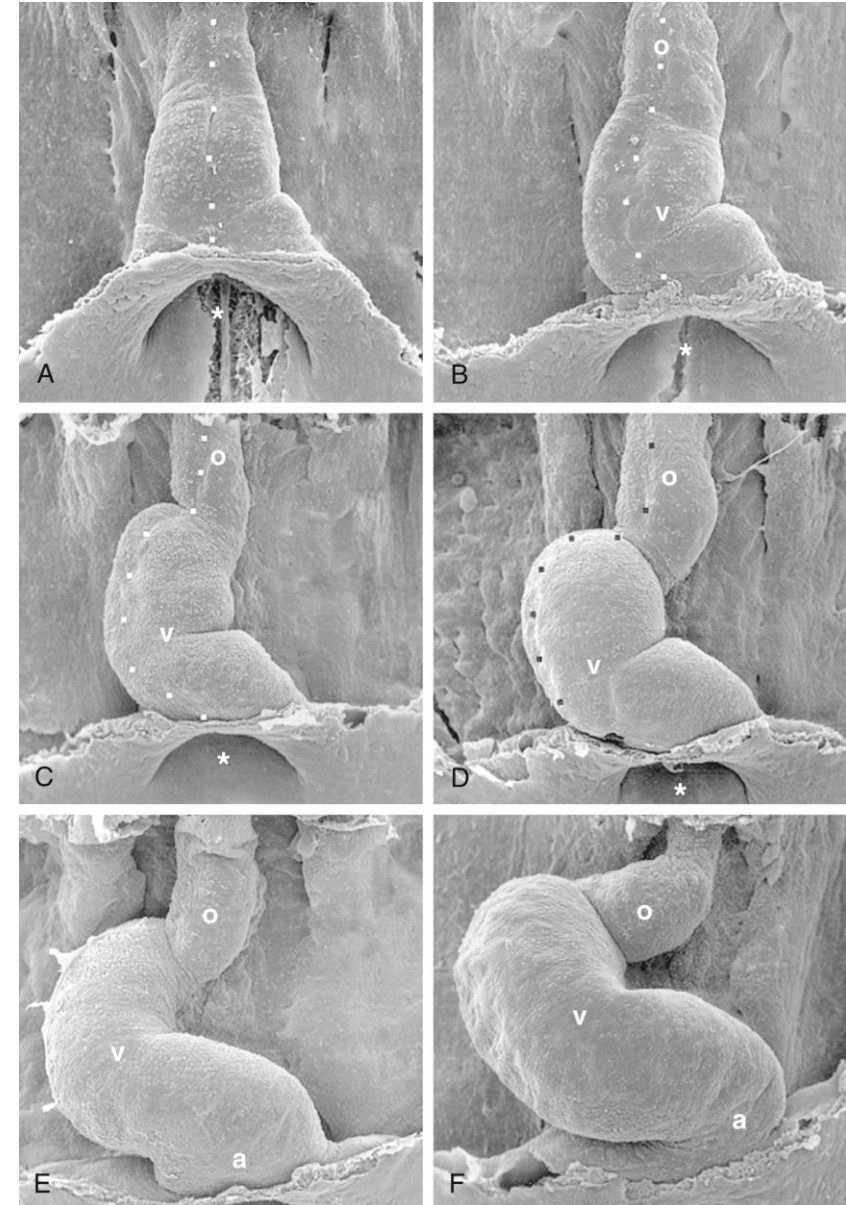
- Primitive endocardial tube is surrounded by splanchnic mesoderm containing myocardial progenitors = **myocardium**
- Developing myocardium deposits an acellular extracellular matrix = **cardiac jelly**
- The **dorsal mesocardium** forms from the second heart field (**SHF**) anchoring the heart tube in place
 - With exception of this dorsal part the heart is suspended in the pericardial cavity
- Cells from the **SHF is added to the cranial portion** of the developing heart tube causing it to elongate
- There's continual expansion of the myocardium
- The primitive heart tube can be differentiated into different portions
 - Outflow tract (prox. Conus arteriosus, dist. Truncus arteriosus which is continuous with aortic sac)
 - Bulbus cordis
 - Ventricle
 - Atrium
 - Sinus venosus



Week 4 – Day 23 – 28 – Cardiac looping

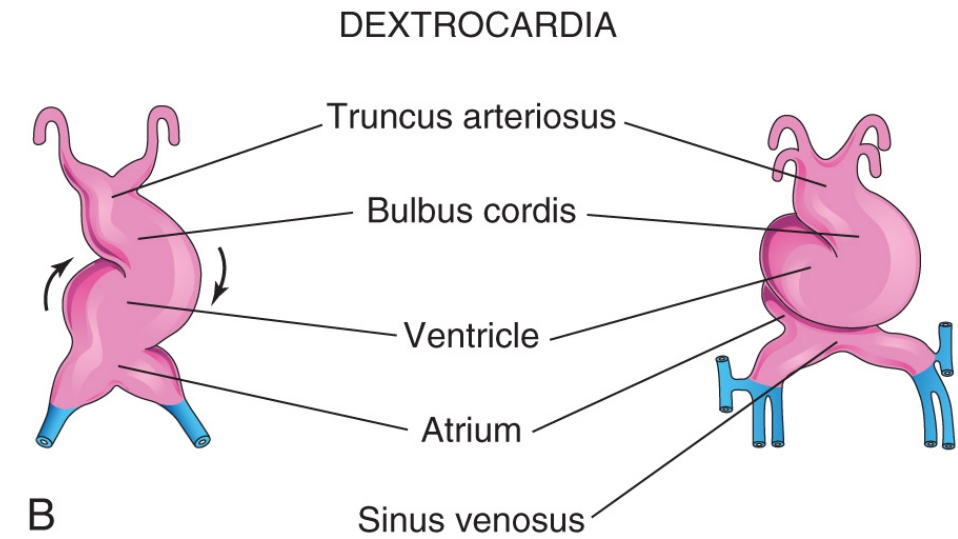
- Day 23 - Primary heart tube elongates and bend into a **C-shaped structure**
 - In addition to bending it also twists
 - Results in the ventral surface becoming the right outer curvature
- The **dorsal mesocardium breaks apart** causing the dorsal side of the tube to become situated in the inner curvature of the C-shaped loop
- With continual elongation the tube becomes **S-shaped**
 - Primitive right ventricle is displaced caudally, ventrally and to the right
 - Primitive left ventricle is displaced to the left
 - Primitive atrium is positioned dorsal and cranially

The heart begins to pump blood on day 24 as looping continues

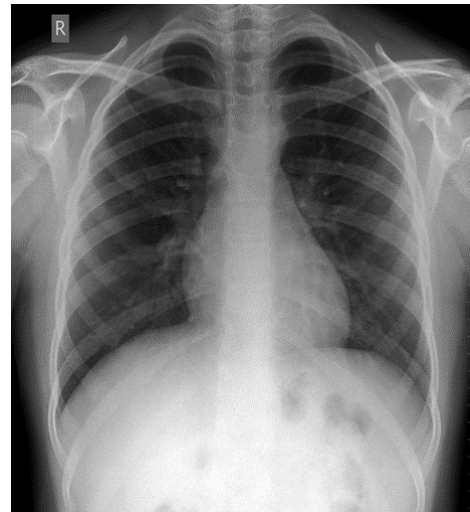


Dextrocardia & situs inversus

- Dextrocardia = Heart tube bends to the left instead of to the right.
- Situs inversus = Dextrocardia + Transposition of the abdominal viscera
- Isolated dextrocardia is usually complicated by severe cardiac defects (e.g., a single ventricle and transposition of the great vessels).
- Situs inversus happens in Primary ciliary dyskinesia = incidence of accompanying cardiac defects is low → Heart functions normally.



Situs inversus



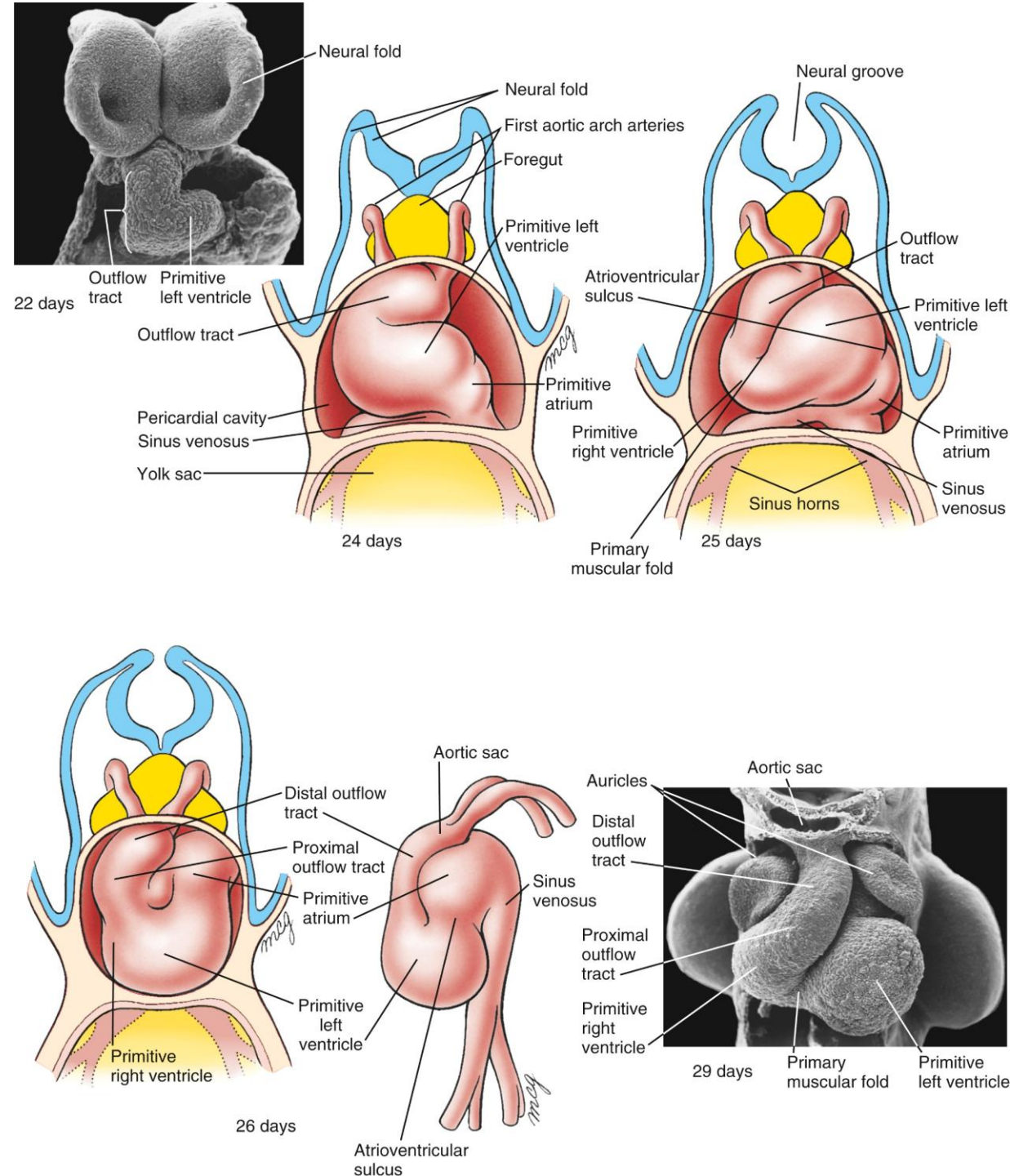
Normal



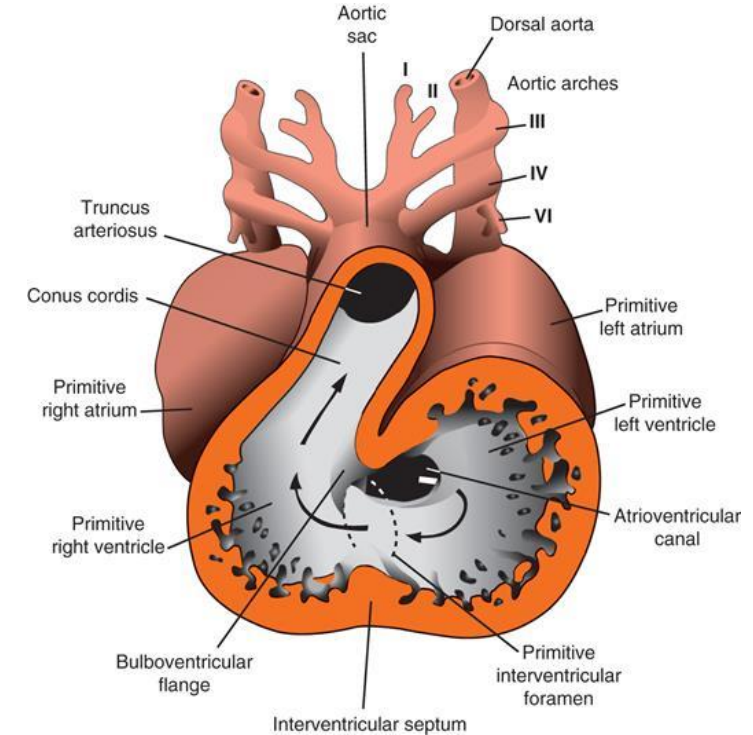
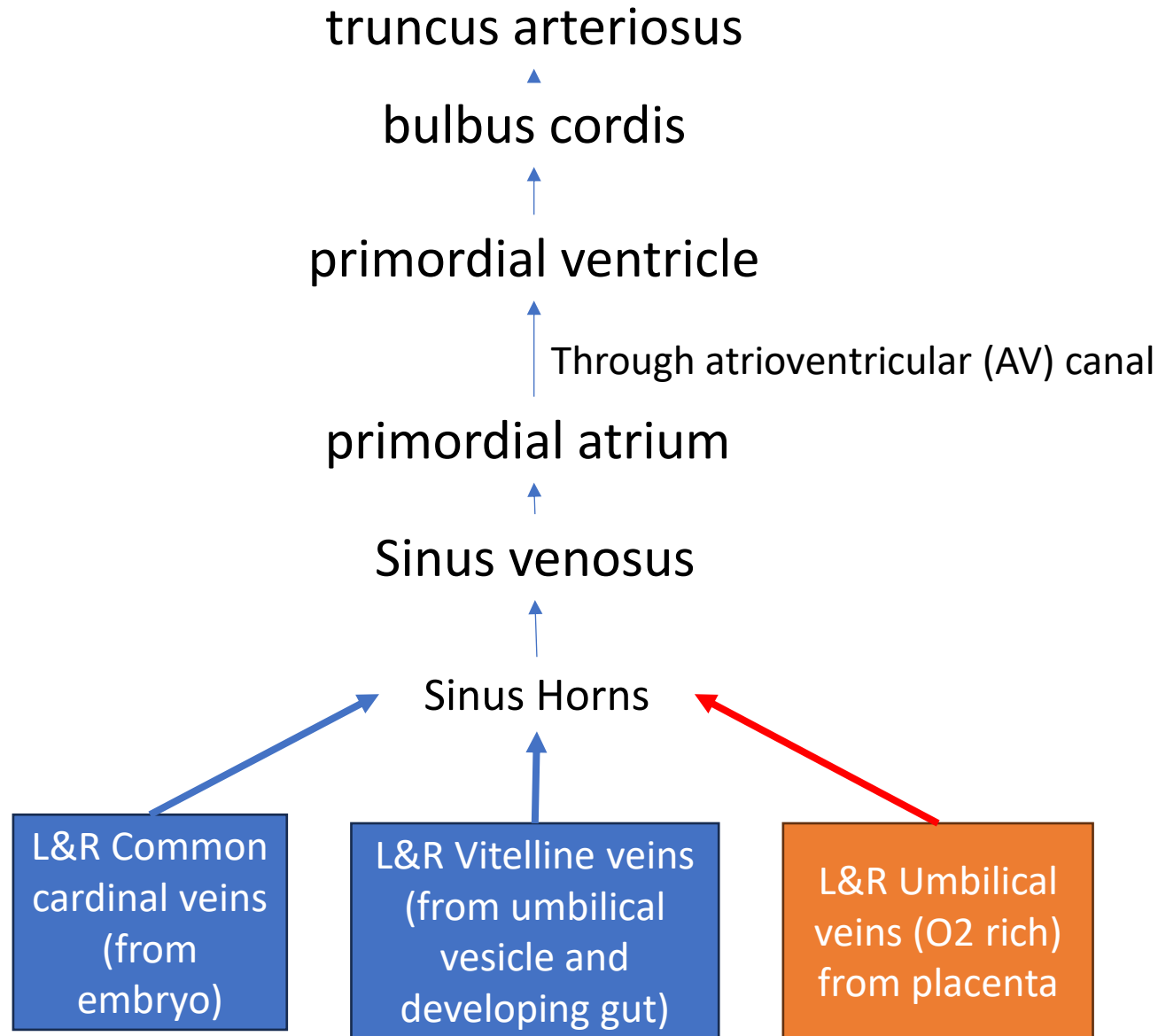
Isolated Dextrocardia

Week 4 –Day 28 and 29

- After completion of the elongation and looping of the heart tube **remodeling continues**
- Day 29 – the outflow tract has positioned itself **ventral to and between** the future left and right atria
- During this process myocardium is continually added
- Continual development occurs over the subsequent weeks and consists of **remodeling and septation** as necessary

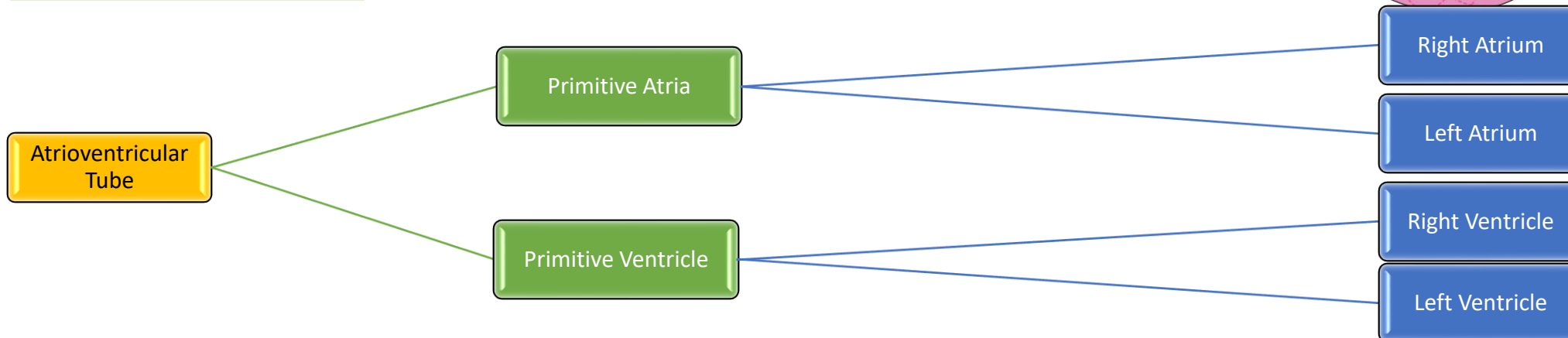
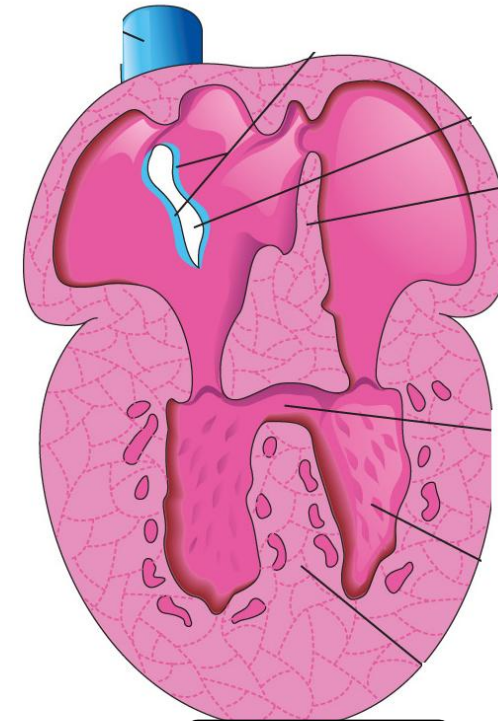
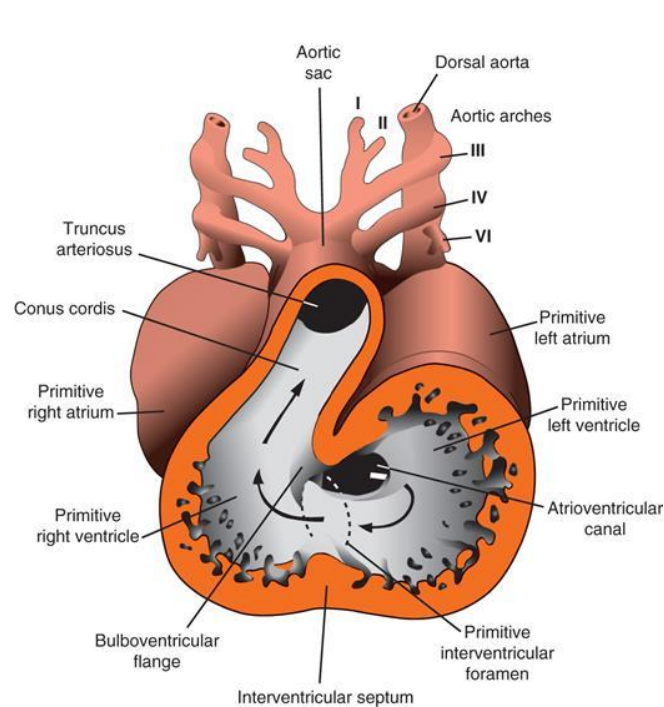


Blood flow to and through the early embryonic heart



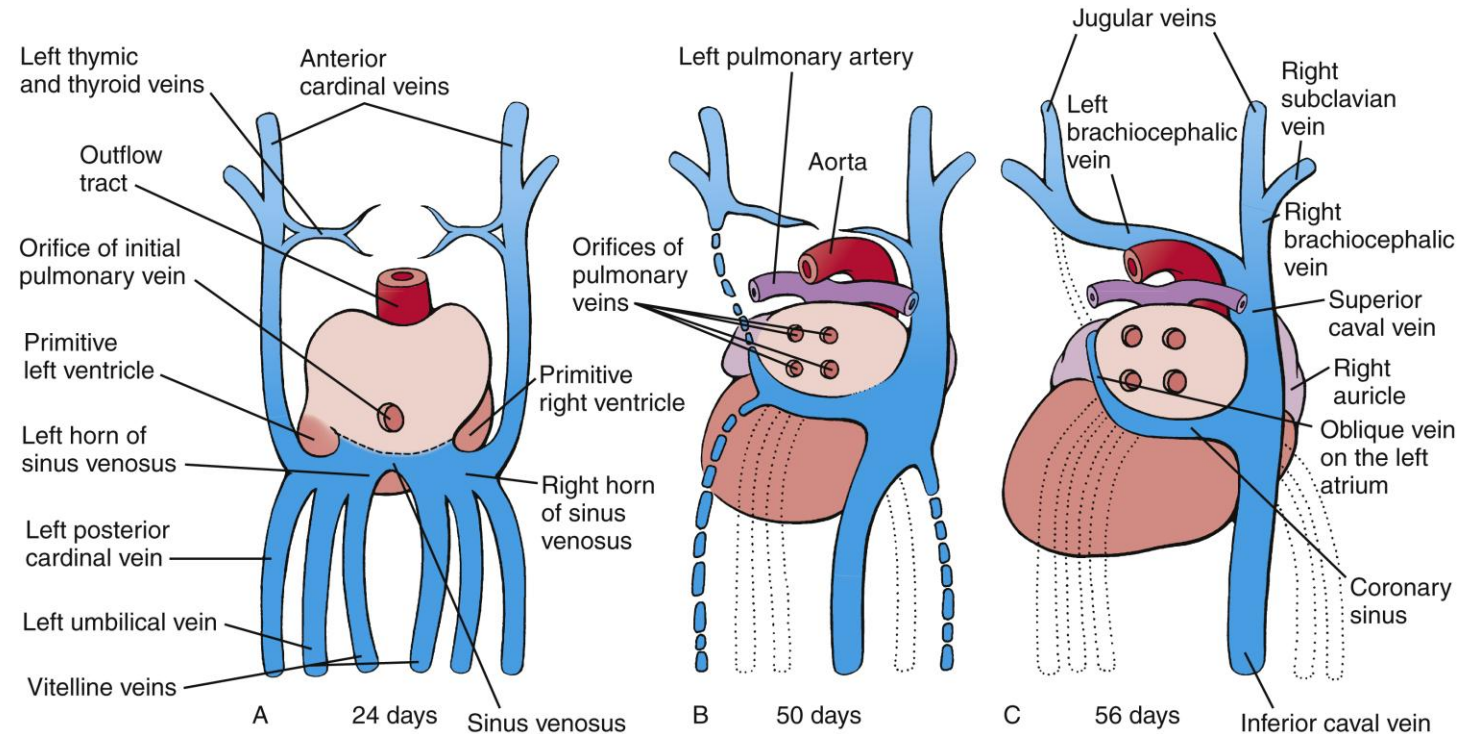
Remodeling and further development

At this early stage the **R. Vent** has exclusive access to the **outflow tract** while the **L. Vent** has exclusive access to the **atrioventricular canal**



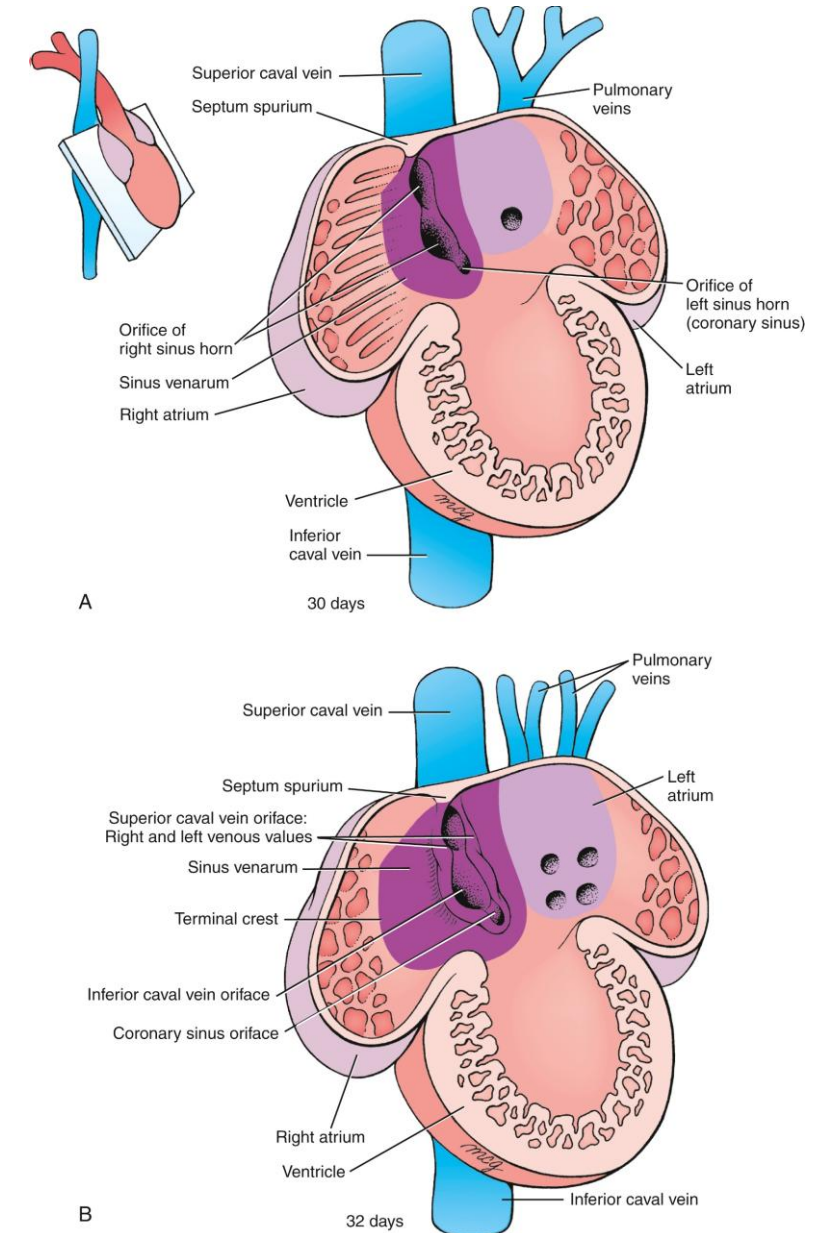
Fate of the sinus horns and sinus venosum

- Venous flow shifts to the right causing progressive enlargement of the **right horn**
 - Anterior cardinal veins increase in size
 - A left to right shunt develops connecting the left and right anterior cardinal veins
 - Results in formation of the superior vena cava
- Right horn receives blood from:
 - Head and neck through **SVC**
 - Placenta and caudal regions of the body through **IVC**
- Left horn regresses and remains only as the coronary sinus



Development of the atria

- Rapid growth of the right sinus horn and caval veins cause the **right side of the sinus venosum to be incorporated into the developing right atrium**
 - Displacing the original atrial wall to the right forming the smooth part = **sinus venarium**
 - Displaces the opening of the coronary sinus to the right
 - The remainder of right atrium (rough part) is derived from primordial atrium
- During septation of the atria the **pulmonary vein in formed in the dorsal mesenchymal protrusion**
 - The pulmonary vein becomes **positioned to the left** as septation continues
 - The pulmonary vein sends **two branches to each lung**
- As the left atrium expands through myocardial growth the main trunk of the pulmonary vein is incorporated into it
 - **Smooth part of the left atrium**

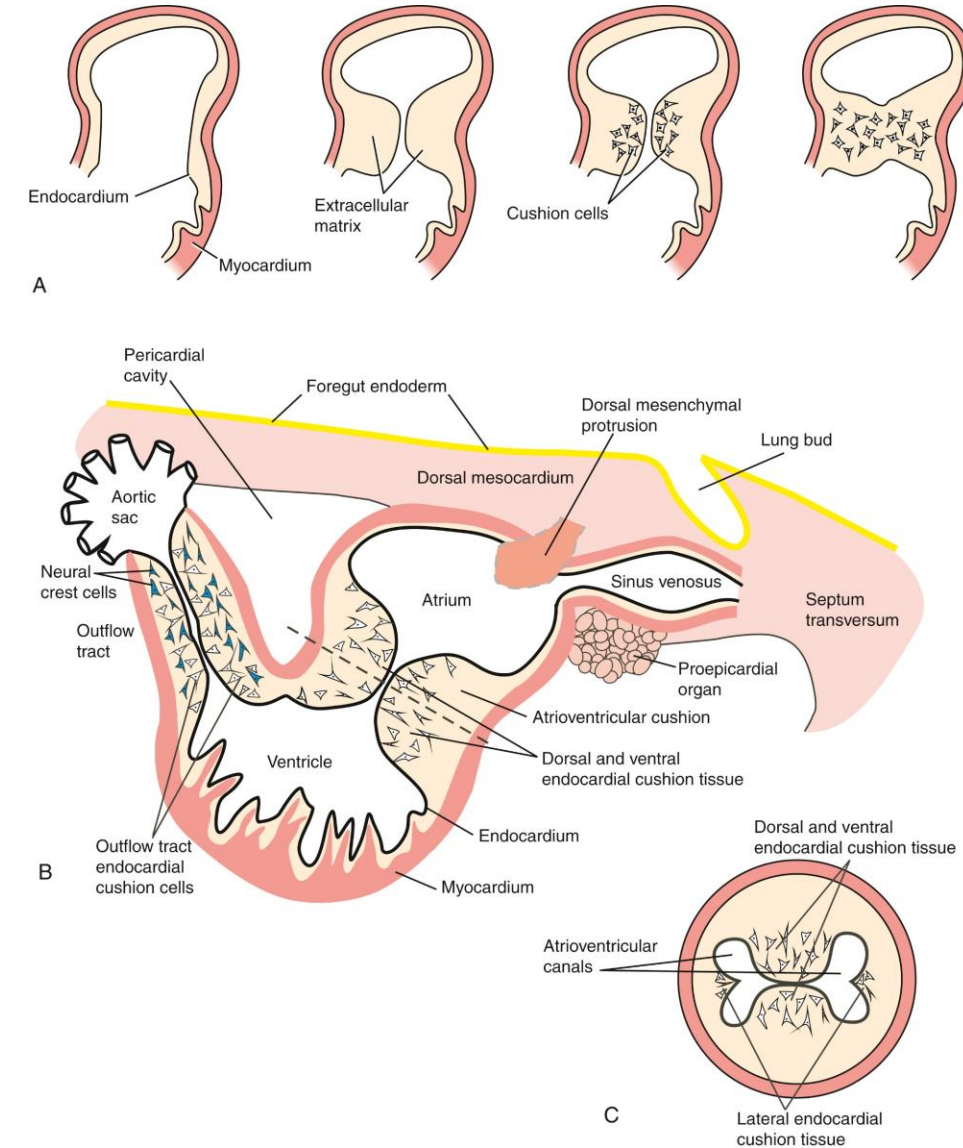


Separation of the ventricles and Atria

Partitioning of Atrioventricular tube

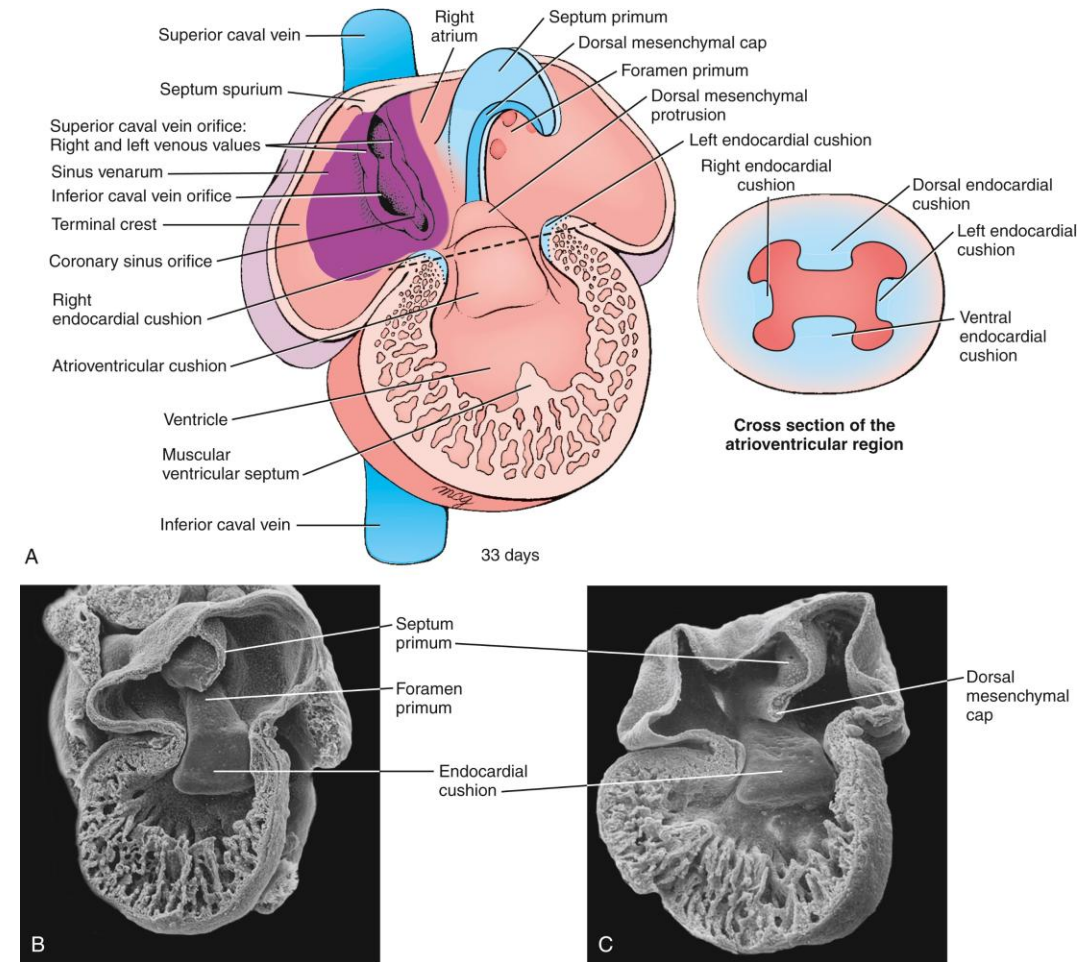
- Cardiac jelly secreted by the myocardium causes the endocardium to bulge inward
- These areas of cardiac jelly become populated with mesoderm derived mesenchyme and neural crest cells = **endocardial cushions**
- AV endocardial cushions form on the dorsal and ventral walls of the AV canal
- Endocardial cushions **grow towards each other and eventually fuse**
- Resulting in separate primordial atrium and primordial ventricle.
- Endocardial cushions act as AV valves
- At the same time and in the same way endocardial cushions develop in the outflow tract
- Repositioning of the ventricles to align with their respective atria also occur during this time*

****the mechanism by which this happens is unclear****



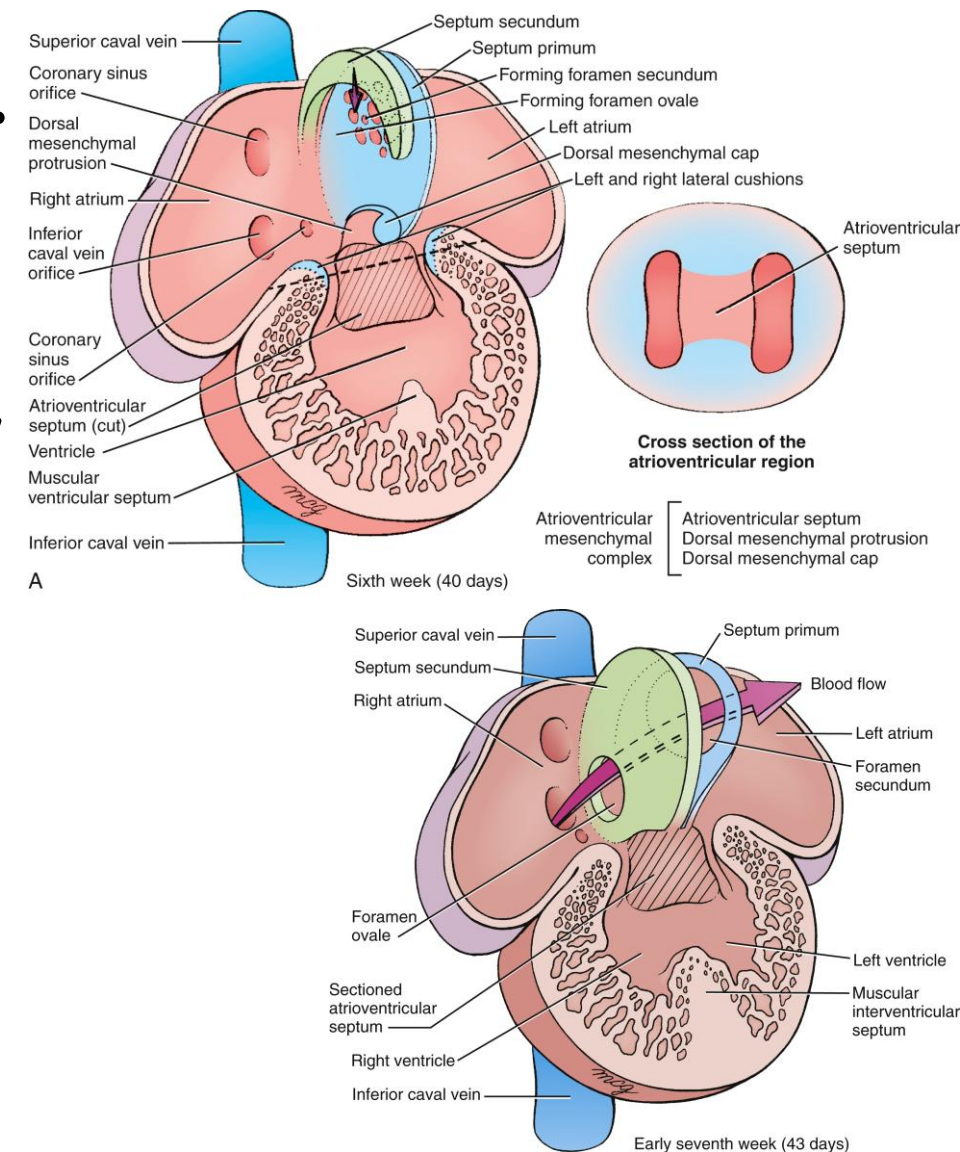
Septation of the atria

- Day 26 - Atrial remodeling is actively ongoing
 - A depression forms along the midline of the atrial roof, beneath the overlying outflow tract
- Day 28 - depression deepens, forming a thin crescent-shaped myocardial wedge = **septum primum**
 - **Extends into the atrium** from the cranial-dorsal wall
 - Causes development of the **foramen primum**
 - On the leading edge the dorsal mesenchymal cap forms, **contains endocardium derived mesenchyme**
 - Cells from the SHF migrate into the atrium forming the dorsal mesenchymal protrusion
 - Adjacent to both the mesenchymal cap and the dorsal endocardial cushion



Septation of the atria cont.

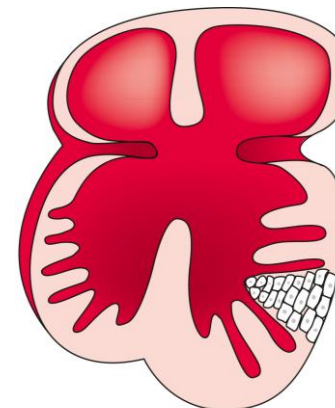
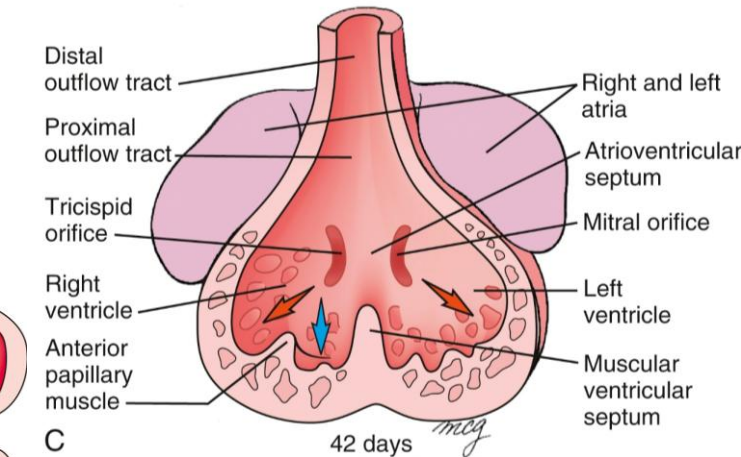
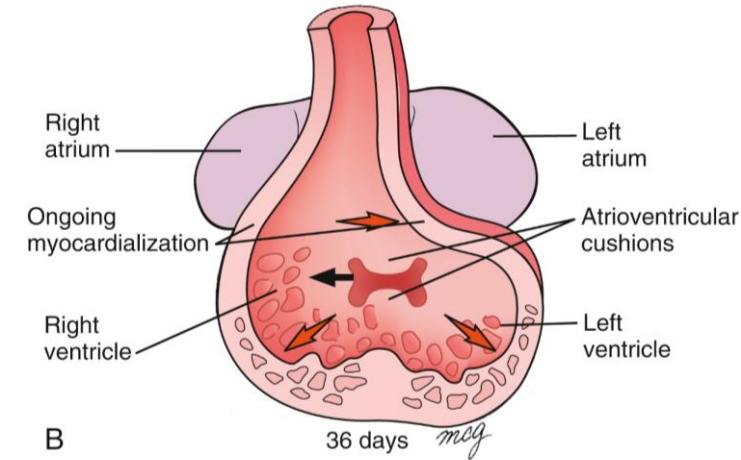
- Septum primum **elongates by differential growth**
 - Simultaneous with fusion of the dorsal and ventral atrioventricular endocardial cushions
- A second crescent shaped septum, thick and muscular, forms on the right adjacent to the septum primum = **septum secundum**
- Fusion of the dorsal mesenchymal cap, dorsal mesenchymal protrusion, and atrioventricular septum = **atrioventricular mesenchymal complex**
 - Separates the atria completely
- Apoptosis in the dorsal region of the septum primum creates perforations = **foramen secundum**
- Septum secundum grows cranial-caudally and dorso-ventrally but does not reach all the way to the endocardial cushion
 - Creates the **foramen ovale**



This arrangement allows blood to pass from right to left through the foramen ovale, then foramen secundum into the left atrium

Week 4/5 - Ventricular septation

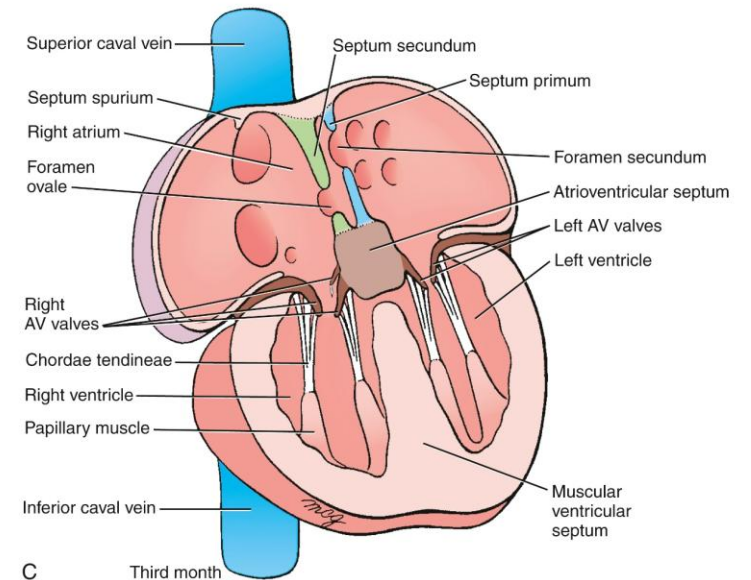
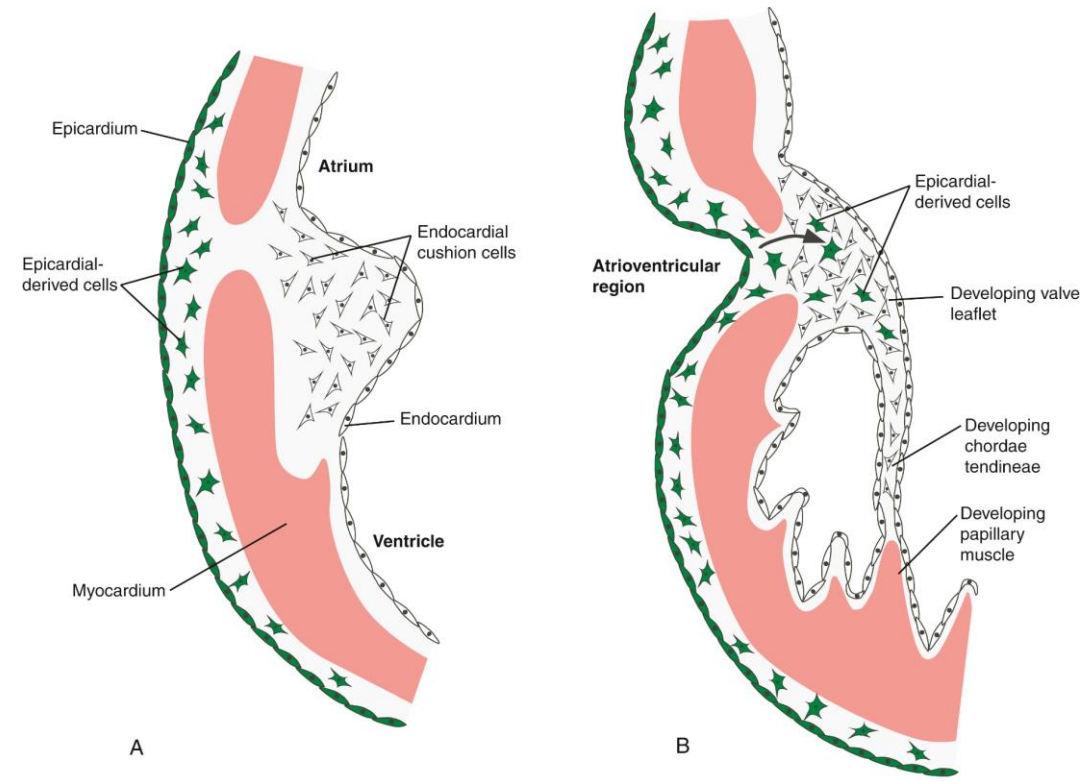
- The **muscular ventricular septum** begins to emerge prominently
 - It is located between the **presumptive right and left ventricular chambers**
 - Grows through addition of cardiomyocytes
 - **Neural crest** cells contribute **membranous** interventricular septum
 - The ventricles **continue expansion** through addition of cardiomyocytes
- The myocardium starts to thicken forming ridges = **trabeculae**
- The septum continues to grow
- **Closure of the interventricular foramen (primary ventricular foramen)** does **not occur until week 8**
 - **Premature closure results in obstruction of the left ventricle** from the **ventricular outflow tract**



Trabeculae and multilayered compact layer

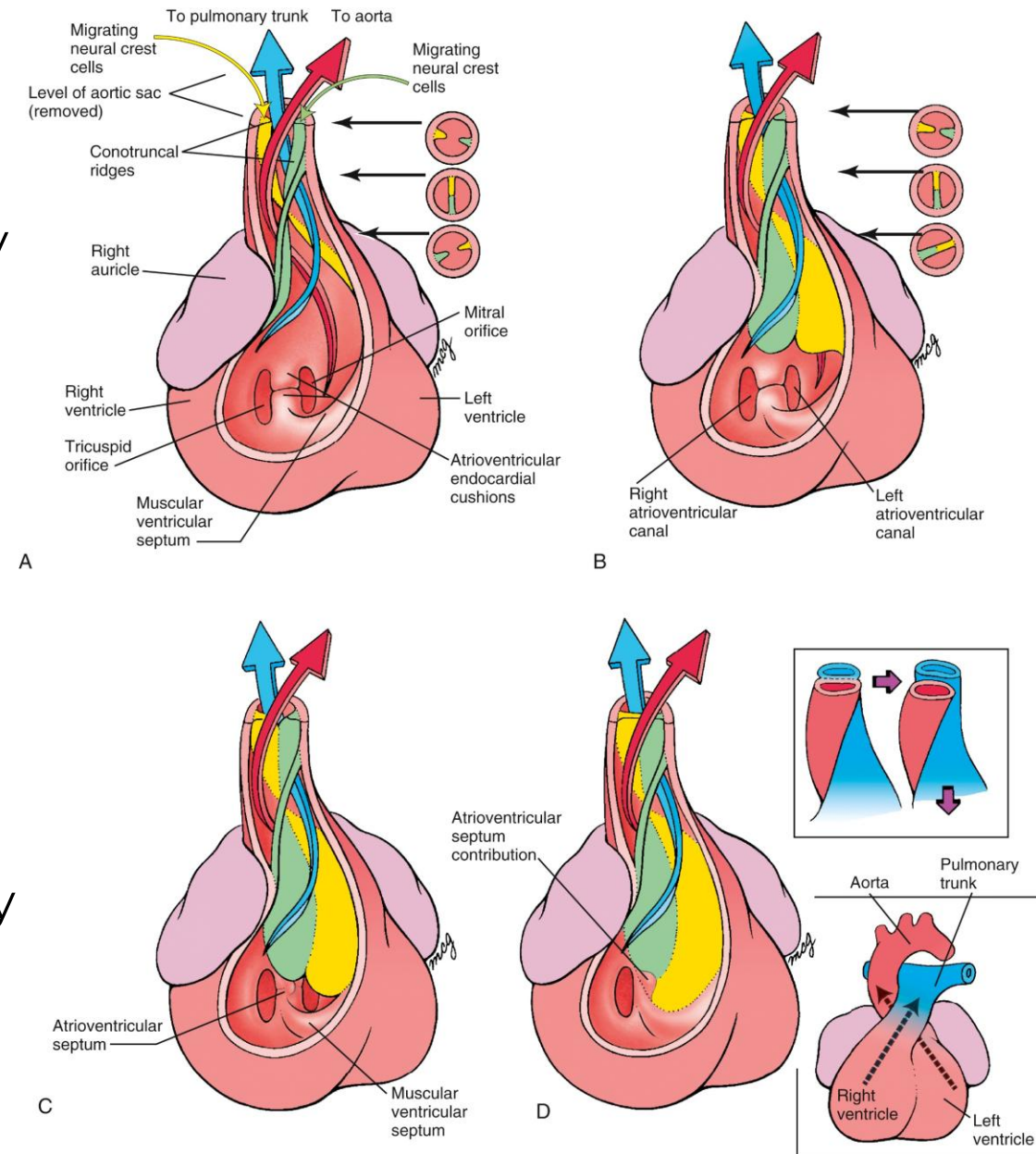
Atrioventricular valve formation

- Leaflets arise from **proliferation and differentiation of the endocardial cushions**
- **Remodeling** and erosion of the ventricular wall **free them**
 - Results in out pockets beneath the valve primordia
 - Thin strands of cells remain connected and result in the formation of the **chorda tendinea and papillary muscles**



Outflow tract separation

- The outflow tract needs to separate in such a way that **each ventricle communicates with its own portion**
- 2 sets of **conotruncal ridges** (left and right) develop along the outflow tract
 - They **are endocardial and neural crest** populated bulges
 - They have **different positions (in a spiral)** along the track
 - They grow towards each other and fuse in a cranial to caudal direction
 - Resulting in a **spiral** structure that completely separates the outflow tract into two
- Most distal part divides because of a **wedge of condensed neural crest** derived mesenchymal tissue that grows down from the aortic sac **between the 4th and 5th aortic arch artery**



Blue and red arrows indicate blood flow from R. and L. ventricles respectively

Development of the semilunar valves

- During development of the outflow tract swellings appear on the main truncus swellings
- In addition, 2 small endocardial cushions develop in opposite quadrants of the distal outflow tract = intercalated cushions
- These are hollowed out to form cavities at the origin of the future vessels

